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CASE STUDY

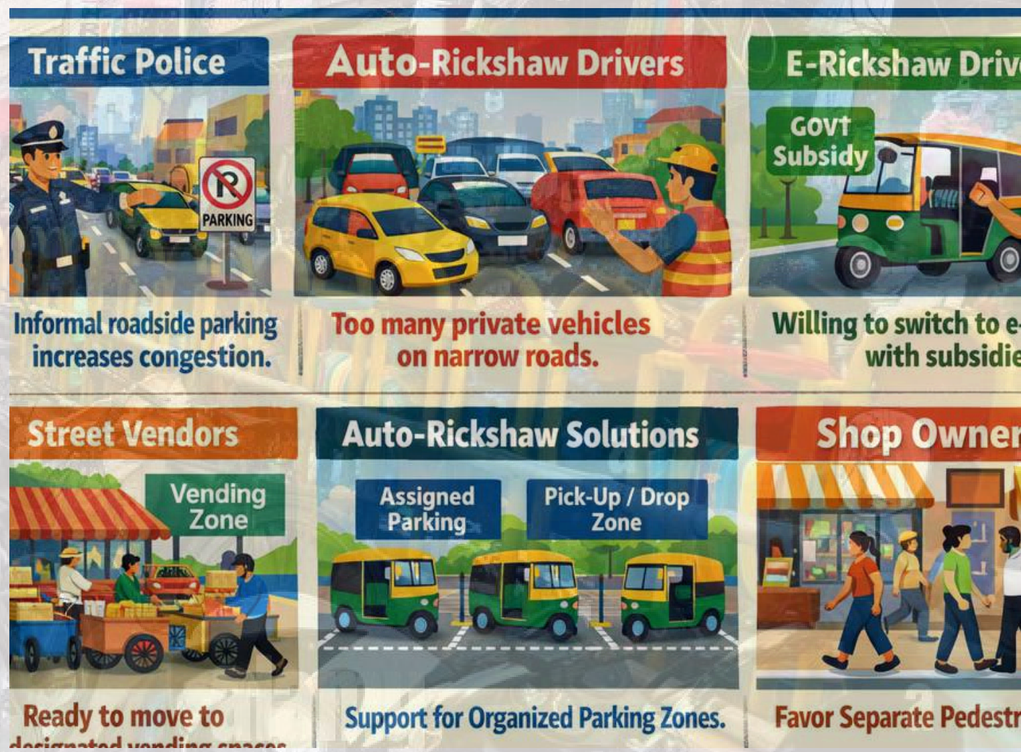
GREEN GANGA CORRIDOR

PREVIEW

- Implementation of AI-based cameras to detect and prevent informal roadside parking, ensuring roads remain fully usable.
- Utilization of existing parking spaces in buildings to shift vehicles away from roadside parking and reduce congestion.
- Introduction of one-way traffic systems on selected roads to improve traffic flow and reduce delays.
- Development of dedicated pedestrian lanes to ensure safe movement of pedestrians and minimize traffic conflicts.
- Creation of the Green Ganga Corridor App to encourage the use of public transport and reduce dependence on private vehicles.
- Introduction of luxury buses to make public transport more comfortable and attractive for private vehicle users.
- Implementation of a travel credit reward system, where passengers earn points for using buses that can be redeemed for discounts on future travel.
- Launch of an EV exchange incentive program, allowing citizens to exchange old vehicles for electric vehicles with discounts and interest-free payment options.

STAKEHOLDER TALKS

- Traffic Police: Identified informal roadside parking as one of the major causes of congestion in busy corridors.
- Auto-rickshaw Drivers: Reported that increased use of private vehicles creates disproportionate traffic volume on narrow roads.
- E-rickshaw Drivers: Expressed willingness to transition to electric rickshaws if supported with government subsidies.
- Street Vendors: Agreed to relocate from roadside areas if provided with designated vending spaces.
- Auto-rickshaw Drivers: Supported the idea of assigned parking and pick-up/drop zones instead of random roadside halting.
- Shop Owners: Favored separate pedestrian lanes, as it would allow customers to walk safely and shop without disrupting traffic



IMPLEMENTATION CHALANGES

- Technology reliability: AI cameras and monitoring systems may face technical failures or require regular maintenance and calibration.
- Public compliance: Drivers may initially resist new rules such as restricted roadside stopping and one-way traffic systems.
- Parking coordination: Partnering with private buildings for parking may require legal agreements, revenue sharing, and operational management.
- Adoption of public transport: Even with incentives, some commuters may continue preferring private vehicles due to habit or convenience.
- Infrastructure readiness: Expansion of EV charging infrastructure and organised parking systems will require phased development and investment.
- Operational enforcement: Effective monitoring and enforcement will require coordination between technology systems and traffic authorities.

IMPROVEMENTS

- Dedicated monitoring centre: Establish a centralised control room to continuously monitor AI cameras and manage traffic operations in real time.
- Public awareness campaigns: Conduct awareness programs to educate citizens about new traffic systems, parking regulations, and the benefits of using public transport.
- Parking lease partnerships: Form multi-year lease agreements with building owners to utilise private parking spaces, ensuring organised and reliable parking availability.
- Cost comparison feature in GGC app: Introduce an app feature that shows commuters the cost difference between using private vehicles and public transport, encouraging informed travel choices.
- Battery swapping for EVs: Implement battery swapping stations where drivers can exchange discharged batteries for fully charged ones, reducing charging time and improving EV usability.

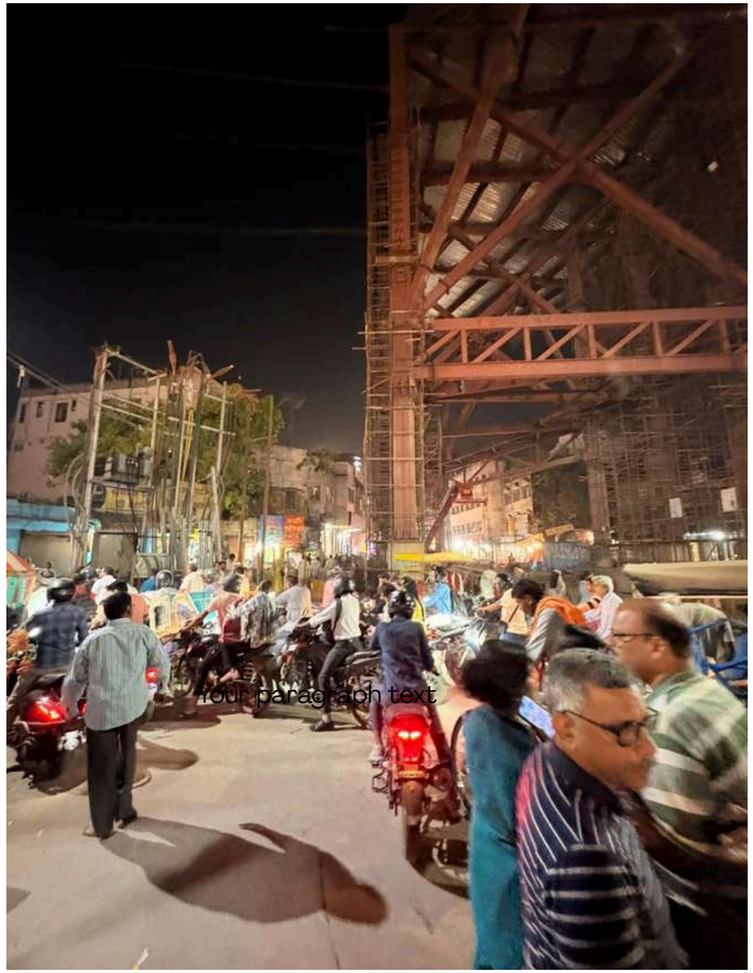
CHARGE SWAPPING

- Battery exchange system: EV drivers can visit battery centres and swap their discharged battery for a fully charged one, eliminating the need for long charging times.
- Time-efficient operation: The swapping process takes only a few minutes, allowing vehicles to remain continuously operational.
- Affordable subscription model: Drivers pay a minimal subscription fee every three months to access the battery swapping service.
- Free monthly swaps: Users receive 3 battery exchanges free every month, making the service more economical and encouraging participation.
- Shared battery responsibility: If a battery is damaged, the user pays only 40% of the replacement cost, while the system covers 60%, thereby reducing financial risk.
- Promoting EV adoption: Faster energy access and reduced operational costs encourage drivers to shift from fuel vehicles to electric mobility

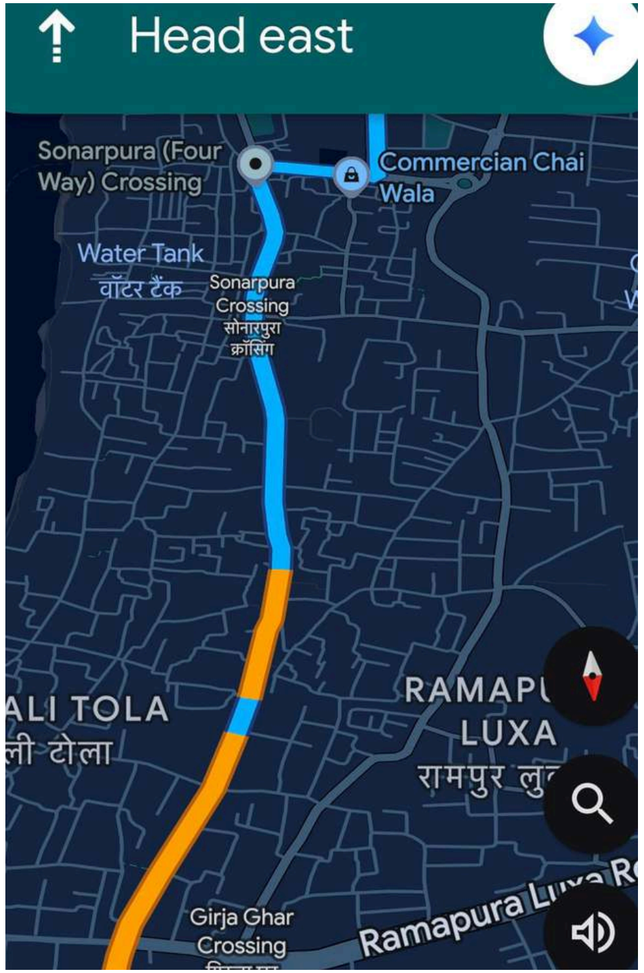
SUMMARY

- Green Ganga Corridor – Integrated Mobility Solution
- AI-driven traffic management to reduce congestion through smart monitoring, adaptive speed control, and one-way corridor systems.
- Organised parking ecosystem using leased private parking spaces integrated with the GGC app to eliminate informal roadside parking.
- Public transport incentives, including premium buses and reward points to encourage commuters to shift from private vehicles.
- Last-mile connectivity through structured e-rickshaw stands at bus stops for smooth and convenient travel.
- EV ecosystem development with subsidies for e-rickshaws and the VoltSwap battery swapping model to enable faster EV adoption.
- People-centred urban design with dedicated pedestrian lanes and regulated vendor spaces to balance mobility and livelihoods.
- Outcome: Reduced congestion, improved mobility efficiency, cleaner transportation, and a scalable model for sustainable urban corridors

THANKS TO OUR STAKEHOLDERS FOR THEIR COOPERATION



GREAT INITIATIVE BY OUR GOVERNMENT FOR THE PADESTRIAN TRAFFIC (ROPEWAYS)



ONE-WAY ROAD CAN EASILY BE MADE ON GOWDOLIA

https://youtu.be/rZCLtCuyW9A?si=NWSFHG8iYRQ3SDEI

ON-GOUND STAKEHOLDER INTERVIEW

VARANASI
OF THE DREAMS
Thank you!!
EV BUS
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